

Geostatistical analysis and conditional simulation for estimating the spatial variability of hydraulic conductivity in the Choushui River alluvial fan, Taiwan

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Abstract:

This work evaluated the spatial variability and distribution of heterogeneous hydraulic conductivity (K) in the Choushui River alluvial fan in Taiwan, using ordinary kriging (OK) and mean and individual sequential Gaussian simulations (SGS). A baseline flow model constructed by upscaling parameters was inversely calibrated to determine the pumping and recharge rates. Simulated heads using different K realizations were then compared with historically measured heads. A global/local simulated error between simulated and measured heads was analysed to assess the different spatial variabilities of various estimated K distributions. The results of a MODFLOW simulation indicate that the OK realization had the smallest sum of absolute mean simulation errors (SAMSE) and the SGS realizations preserved the spatial variability of the measured K fields. Moreover, the SAMSE increases as the spatial variability of the K field increases. The OK realization yields small local simulation errors in the measured K field of moderate magnitude, whereas the SGS realizations have small local simulation errors in the measured K fields, with high and low values. The OK realization of K can be applied to perform a deterministic inverse calibration. The mean SGS method is suggested for constructing a K field when the application focuses on extreme values of estimated parameters and small calibration errors, such as in a simulation of contaminant transport in heterogeneous aquifers. The individual SGS realization is useful in stochastically assessing the spatial uncertainty of highly heterogeneous aquifers. Copyright © 2004 John Wiley & Sons, Ltd.

KEY WORDS spatial variability; hydraulic conductivity; ordinary kriging; sequential Gaussian simulation; MODFLOW

INTRODUCTION

Spatial distributions of the properties of natural aquifers, such as hydraulic conductivity (K), often exhibit high heterogeneity. However, in a field investigation, only a small fraction of *in-situ* data can be analysed, owing to time and cost constraints. The sparse measured data contain a considerable degree of uncertainty. Geostatistics commonly have been used to estimate the spatial variability and distribution of field data with uncertainty. A semivariogram (or variogram) is used to analyse spatial variation structures, and kriging, a best linear unbiased estimator (BLUE), then uses measured data to estimate regional variables at unsampled locations by analysing the spatial variability. Previous research on evaluating the spatial variability and distribution of groundwater parameters includes Delhomme (1979), Hoeksema and Kitanidis (1985), Woodbury and Sudicky (1991), Hess *et al.* (1992), Turcke and Kueper (1996), Fabbri (1997), Nakaya *et al.* (2002) and Liu *et al.* (2002).

One process involved in the kriging technique is to minimize the variance of the prediction error, which requires smoothing the variability of observed values (Eggleson *et al.*, 1996; Lin *et al.*, 2000). The smoothing effect increases with a nugget effect (Goovaerts, 1997). Conditional simulations, which reproduce

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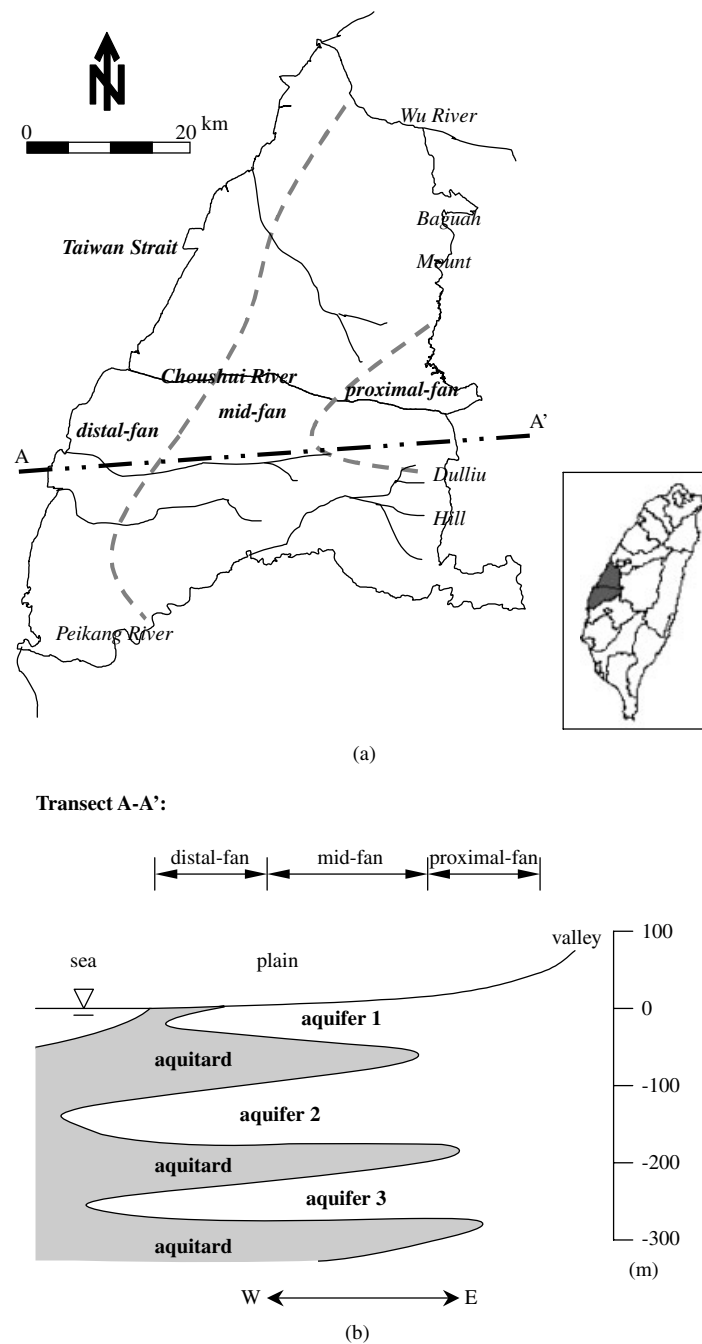


Figure 1. Location and conceptual hydrogeological profile of the Choushui River alluvial fan, Taiwan

certain statistical characteristics of measured data to simulate spatial distributions, can overcome the above shortcoming. Istok and Pautman (1996), Eggleston *et al.*, (1996), Scholl (2000), Lin *et al.* (2000) and Feyen *et al.* (2001) used conditional simulations to reproduce spatial distributions of groundwater quality, K , and transmissivity.

Numerical simulations usually are applied in the hydrogeological field for solving groundwater flow and transport problems, based on their physical and chemical processes. Geostatistical approaches are used frequently to determine hydrogeological parameter distributions as inputs in numerical models (Boman *et al.*, 1995; Poeter and McKenna, 1995; Eggleston and Rojstaczer, 1998; Wheater *et al.*, 2000; Feyen *et al.*, 2001). Additionally, some groundwater modelling packages include a convenient user interface for interpolating the relevant parameter fields using various geostatistical methods, such as GMS (U.S. Department of Defense, 1998) and UNCERT (Wingle *et al.*, 1999). However, simulated results concerning groundwater flow and contaminant transport can vary greatly, because various spatial interpolation algorithms are applied for K fields.

The objective of this study is to examine the spatial variability of heterogeneous K fields in the Choushui River alluvial fan, Taiwan. Ordinary kriging (OK) and conditional simulation, mean and individual sequential Gaussian simulations (SGS), were used to estimate the spatial variability and distribution of the K field. A baseline flow model with locally averaged, measured K data was used to calibrate pumping and recharge rate parameters. Subsequently, simulated groundwater heads for different K realizations determined by the geostatistical approaches were compared with historically measured groundwater heads. Global/local errors between simulated and measured heads were analysed to assess the spatial variabilities of various estimated K distributions. The estimated K fields are critical for analysing problems of land subsidence and contaminant transport, and for delineating capture zones (Wheater *et al.*, 2000; Liu *et al.*, 2001; Feyen *et al.*, 2001). The results of this study are useful for establishing a sound water management plan to prevent further drawdown of groundwater levels, land subsidence and aquifer contamination in the Choushui River alluvial fan, Taiwan.

STUDY AREA

The Choushui River alluvial fan is located on the western coast of Taiwan and enclosed by the Taiwan Strait to the west, the Wu River to the north, the Dulliu Hill and Baguah Mount to the east, and the Peikang River to the south, as depicted in Figure 1a. The alluvial fan has an area of around 1800 km² and is partitioned primarily into the proximal-fan, mid-fan and distal-fan areas. Unconsolidated sediments that underlie the alluvial fan contain abundant groundwater and are of late Quaternary age. The proximal-fan is a major region for aquifer recharging. The long-term average amounts of groundwater pumped and recharged from 1970 to 1990 were 1.33 and 1.02 billion m³/year, respectively. These statistics indicate that a substantial amount of groundwater was overdrafted from the aquifers (Central Geological Survey, 1999). Thus, groundwater levels dropped markedly from 1970 to 1990 but recovered slightly after 1991.

Drilling and stratigraphical analyses of the subsurface geology were conducted from 1992 to 1998 in order to understand further the hydrogeological characteristics of the alluvial fan. A new groundwater monitoring network, including 77 hydrogeological investigation stations and 188 groundwater monitoring wells in different aquifers (Taiwan Sugar Company, 1997), was quickly established. Subsurface hydrogeological analysis to a depth of approximately 300 m is divided into six overlapping sequences, including three marine sequences and three non-marine sequences, in the distal-fan and mid-fan areas (Central Geological Survey, 1999). Generally, non-marine sequences, with coarse sediment, ranging from medium sand to gravel with high permeability, are considered to be aquifers, whereas marine sequences, with fine sediment, ranging from clay to fine sand with low permeability, are regarded as aquitards (Figure 1b). The hydrogeological proximal-fan formation, which comprises mainly gravel and sand, serves as an unconfined aquifer. Aquitards mainly are located in the distal-fan and mid-fan areas, but not in the proximal-fan area. The upper, intermediate and lower aquifers are labelled 'aquifer 1', 'aquifer 2' and 'aquifer 3', which contain 39, 59 and 43 monitoring wells, respectively.

Data derived from pumping tests indicate that the observed K fields are large in the proximal-fan and small in the distal-fan and at the southern and northern boundaries (Taiwan Sugar Company, 1997) (Figure 2). Freeze (1975) noted that subsurface K values frequently follow a log-normal distribution. Therefore, the natural logarithm, $\ln(K)$, of the measured K is taken. Statistical analyses show that the measured $\ln(K)$ ranges

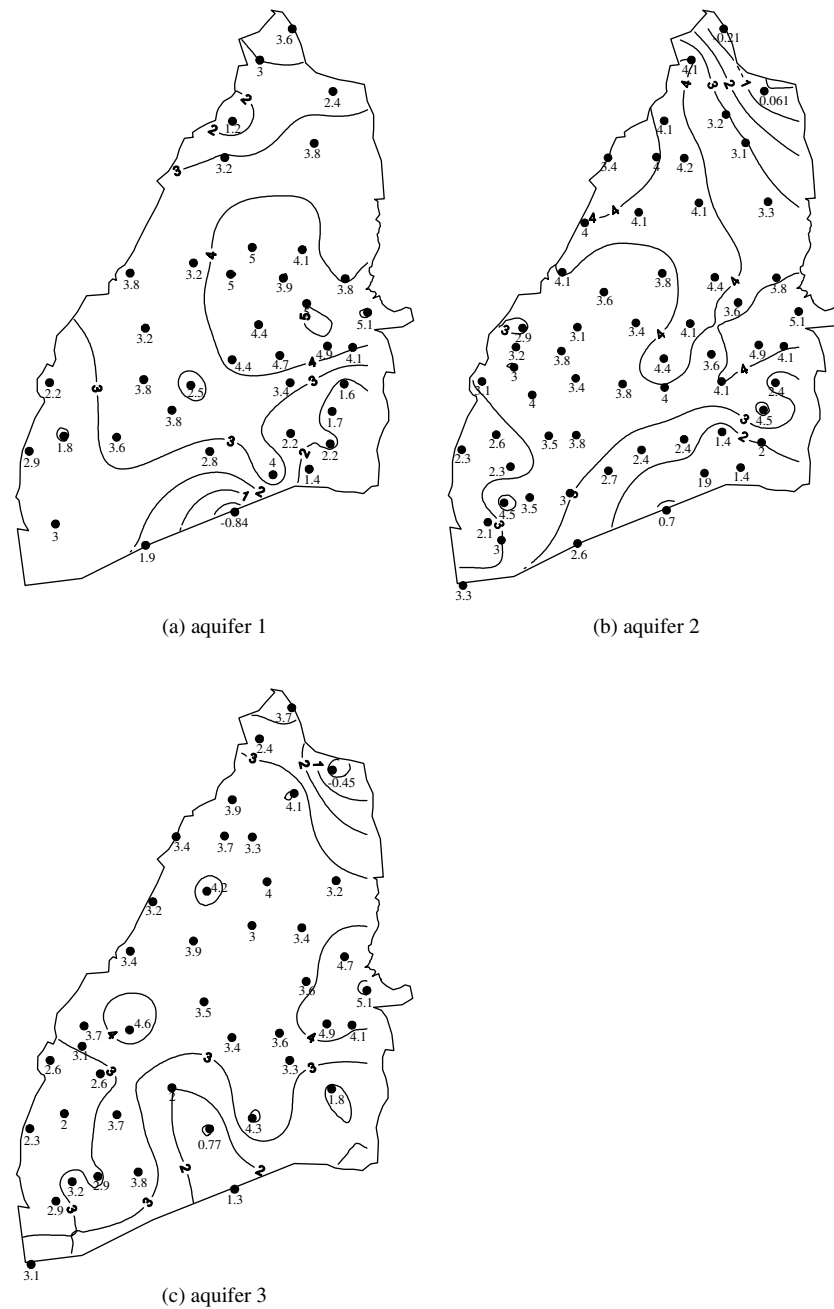


Figure 2. Measured $\ln(K)$ in (a) aquifer 1, (b) aquifer 2 and (c) aquifer 3 (in $\ln(\text{m/day})$)

from 5.12 to $-0.84 \ln(\text{m/day})$ in the three aquifers (Table I). The mean value of $\ln(K)$ varies slightly among the aquifers and is less than the median. Although the $\ln(K)$ distributions are less skewed than the K distributions, the skewness of $\ln(K)$ still ranges from -0.78 to -1.23 . The $\ln(K)$ is checked by the goodness-of-fit to a normal distribution using a chi-square test. The measured $\ln(K)$ of the aquifers is normally distributed at a 5% level of significance.

Table I. Descriptive statistics of measured $\ln(K)$

Statistics	Aquifer 1	Aquifer 2	Aquifer 3
Mean ($\ln(\text{m/day})$)	3.22	3.24	3.24
Median ($\ln(\text{m/day})$)	3.37	3.44	3.39
Standard deviation ($\ln(\text{m/day})$)	1.29	1.09	1.07
Skewness	-0.78	-1.18	-1.23
Minimum ($\ln(\text{m/day})$)	-0.84	-0.21	-0.45
Maximum ($\ln(\text{m/day})$)	5.12	5.12	5.12

METHODS

When certain parameters are inversely calibrated in a numerical model, a zonation distribution is usually applied to scale up calibration parameters (Yeh, 1986; Wen and Gómez-Hernández, 1996; Gau *et al.*, 1998; Ting *et al.*, 1998; Vassolo *et al.*, 1998). In this study, a zonation distribution was adopted to construct spatial patterns of pumping and recharge rates, and locally average measured K fields in a baseline flow model. The computer codes gamv, kt3d and gsim in GSLIB (Deutsch and Journel, 1998) were utilized to generate experimental variograms, OK and SGS, respectively. The estimated K values in each aquifer are imposed on a grid of 27×40 cells, and the size of each cell is $2 \text{ km} \times 2 \text{ km}$. The estimated outcomes are analysed using the SPSS statistical package (SPSS Inc., 1998). Once the baseline flow model was calibrated, the K fields by OK, mean SGS and individual SGS were imported into a flow model, MODFLOW (McDonald and Harbaugh, 1988), for numerical simulation. The differences between measured and simulated heads were

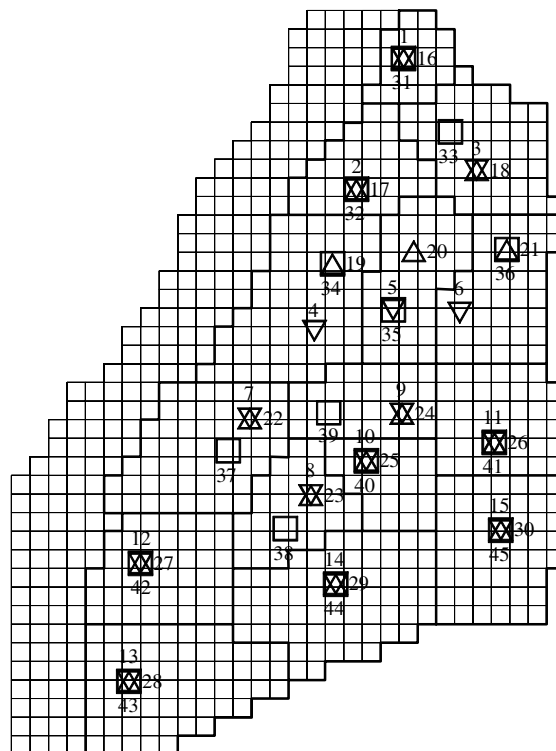


Figure 3. Plan view of zonation distributions in the Choushui River alluvial fan (∇ , monitoring well in aquifer 1; Δ , monitoring well in aquifer 2; $\square$, monitoring well in aquifer 3)

analysed in order to evaluate the accuracy and applicability of different geostatistical estimates. The adopted geostatistical approaches are discussed as follows.

Variogram analysis

The assumption of second-order stationary generally is applied in the geostatistical spatial analysis. A variogram quantifies the spatial variability of random variables between two sites. In practice, an experimental variogram, $\gamma(\mathbf{h})$, calculates pairs of data points separated by a vector, $\mathbf{h}$. The experimental variogram is then fitted by a theoretical model, such as a spherical, exponential, Gaussian or power model, to determine three parameters, including a nugget effect, sill and range. However, variograms also measure the spatial variability along a certain direction given an angle of tolerance. An omnidirectional variogram is used to analyse spatial variation structures herein because the pairs of data are not enough to enable multidirectional analyses to be performed.

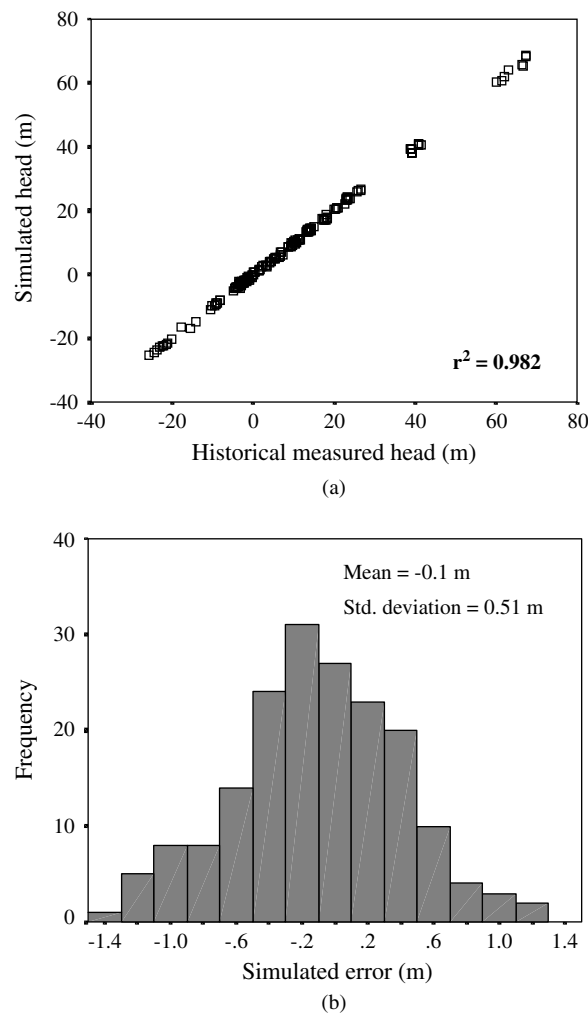


Figure 4. (a) Scatter plot of simulated against historically measured groundwater heads. (b) Frequency histogram of simulation errors

Ordinary kriging

Ordinary kriging is a commonly used geostatistical method for constructing K fields as inputs to groundwater flow and transport models. Ordinary kriging is a linear weighted-average technique, which is unbiased for expected values of estimators and random variables, and minimizes the estimation error variance. A kriging estimator not only provides an estimate but also determines an estimation error variance, a kriging variance, which quantifies the uncertainty in an estimate at each location. This study considers estimated values in each grid cell, but does not include kriging variances. Isaaks and Srivastava (1989) thoroughly discussed ordinary kriging.

Sequential Gaussian simulation

Stochastic simulation is the process of building alternative, equally probable, high-resolution models of a spatially distributed random variable. The simulation is 'conditional' if the resulting realizations honour measured data at their locations.

Sequential Gaussian simulation creates realizations of normal random variables and performs a Gaussian transformation of the available measurements. A simulated value at a visited point is randomly drawn from the conditional cumulative distribution function, defined by the kriging mean and variance, based on

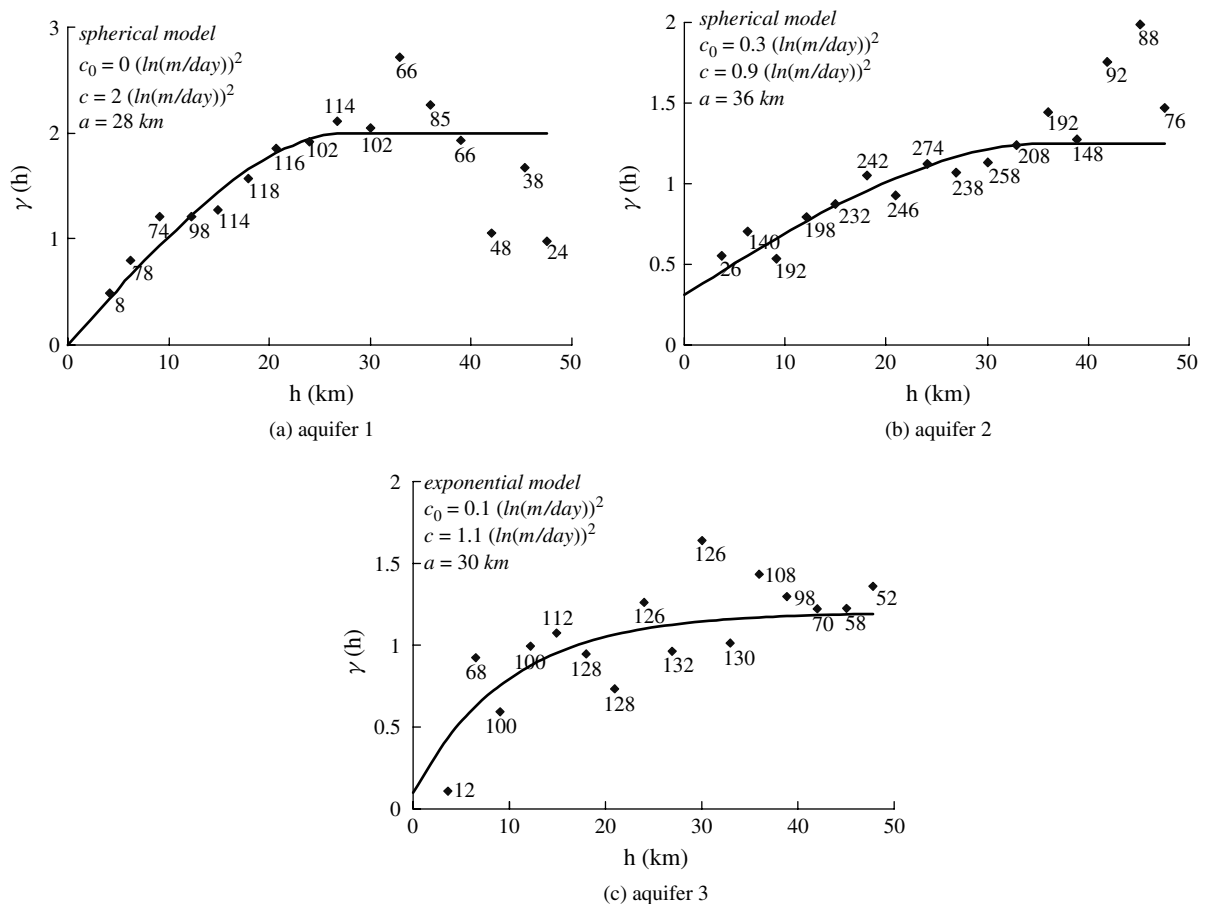


Figure 5. Best-fit theoretical model of experimental variograms and their fitting parameters in (a) aquifer 1, (b) aquifer 2 and (c) aquifer 3. The value below the data point specifies the number of pairs used in the experimental variograms

neighbourhood values. At a new randomly visited point, the simulated value is conditional on the original data and previously simulated values. Finally, the simulated normal values are transformed back into the simulated values for the original variables. The process is repeated until all points are simulated for each realization throughout the grid (Deutsch and Journel, 1998).

Each SGS realization varies greatly in space (Lin *et al.*, 2000), although they condition on observed data. The treatment of multiple realizations significantly differs according to the aims of the study. The treatment typically is divided into two categories. Either a mean procedure that involves multiple realizations is desirable to obtain a realization for a completely deterministic evaluation (mean SGS realization) (Eggleston *et al.*, 1996) or several realizations of conditional simulations are individually performed for a stochastic evaluation (individual SGS realization) (Boman *et al.*, 1995; Poeter and McKenna, 1995; Scholl, 2000). The mean SGS realization and the individual SGS realization are adopted herein to determine the SGS realizations

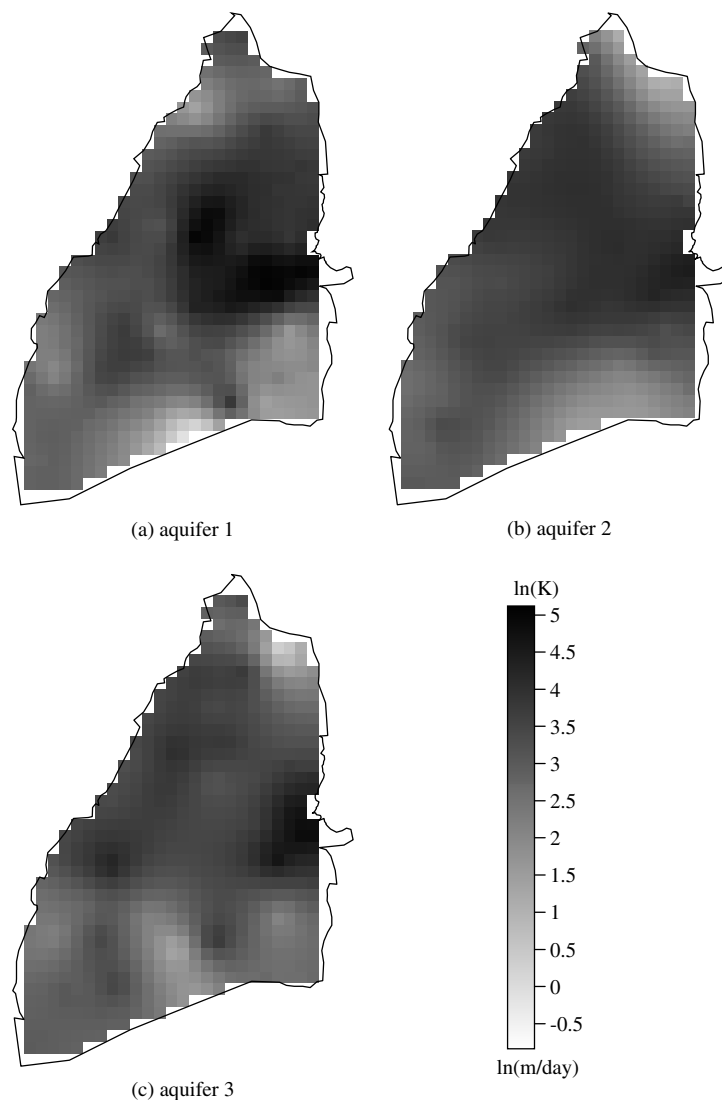


Figure 6. OK realization of $\ln(K)$ in (a) aquifer 1, (b) aquifer 2 and (c) aquifer 3

of K . A SGS realization of the smallest simulated errors is of interest and examined using numerous flow simulations.

RESULTS AND DISCUSSION

Model calibration

The simulation period was from February 1998 to February 1999, and was partitioned into four small periods to represent seasonal fluctuations of groundwater levels between dry seasons (February to April and November to January) and rainy seasons (May to July and August to October). Groundwater levels in January 1998 were set as initial conditions. Dirichlet boundary conditions were imposed on the coast of aquifer 1 and on the easternmost aquifers. The other boundaries were assigned to a zero-flow. Owing to a lack of reliable observed records, the simulation was inversely calibrated to obtain the zonation distributions of the pumping and recharge rates. The modelling domain was divided into 15 zones for each aquifer, according to hydrogeological characteristics, so the three aquifers included a total of 45 zones (Figure 3). In a baseline flow model, a K field within each zone was constructed by determining a geometrical average of local measured data (Wen and Gómez-Hernández, 1996). Net recharge rates were imposed on the uppermost aquifer. Pumping rates were assigned to the two remaining aquifers. Each zone included a monitoring well, from which historically measured groundwater levels were taken for calibration.

The pumping and recharge rates were inversely adjusted to reduce the difference between simulated and measured heads, and the calibrated errors were expected to fall within a tolerance of 1.5 m for each monitoring well (Gau *et al.*, 1998). If calibrated pumping and recharge rates closely reflect realistic situations, then seasonal simulation errors will be relatively small and temporally independent, allowing seasonal simulation errors to be averaged. The sum of absolute mean simulation errors, SAMSE, was calculated for various K distributions, as specified in Equation (1).

$$\text{SAMSE} = \sum_{i=1}^{45} |\bar{e}_i| \quad (1)$$

where,

$$\bar{e}_i = \frac{1}{4} \sum_{j=1}^4 e_{ij}$$

$$e_{ij} = \hat{h}_{ij} - h_{ij}$$

$\bar{e}_i$ is the mean simulation error in the i th zone, e_{ij} is the simulation error during the j th simulation period in the i th zone, $\hat{h}_{ij}$ is the simulated head during the j th simulation period in the i th zone and h_{ij} is the measured head during the j th simulation period in the i th zone.

The calibrated water budget is the net recharge rate of 89 million m^3/year , which is consistent with an average rise in the observed groundwater level of 0.8 m during the period of simulation. Accordingly, the calibrated model reasonably reflects a realistic groundwater flow (Figure 4a), and the simulation errors are almost normally distributed (Figure 4b). Additionally, the simulation errors are insensitive to the simulation period, but not to the zone, based on the analysis-of-variance test at a 5% level of significance. The calculated SAMSE is 17.6 m in the calibration of MODFLOW simulation.

Experimental variogram analysis

Fitting a variogram model is inevitably subjective (Woodbury and Sudicky, 1991). Employing a least squares method to fit a variogram generates a best-fit model with the lowest fitting errors (Cressie and Hawkins, 1980;

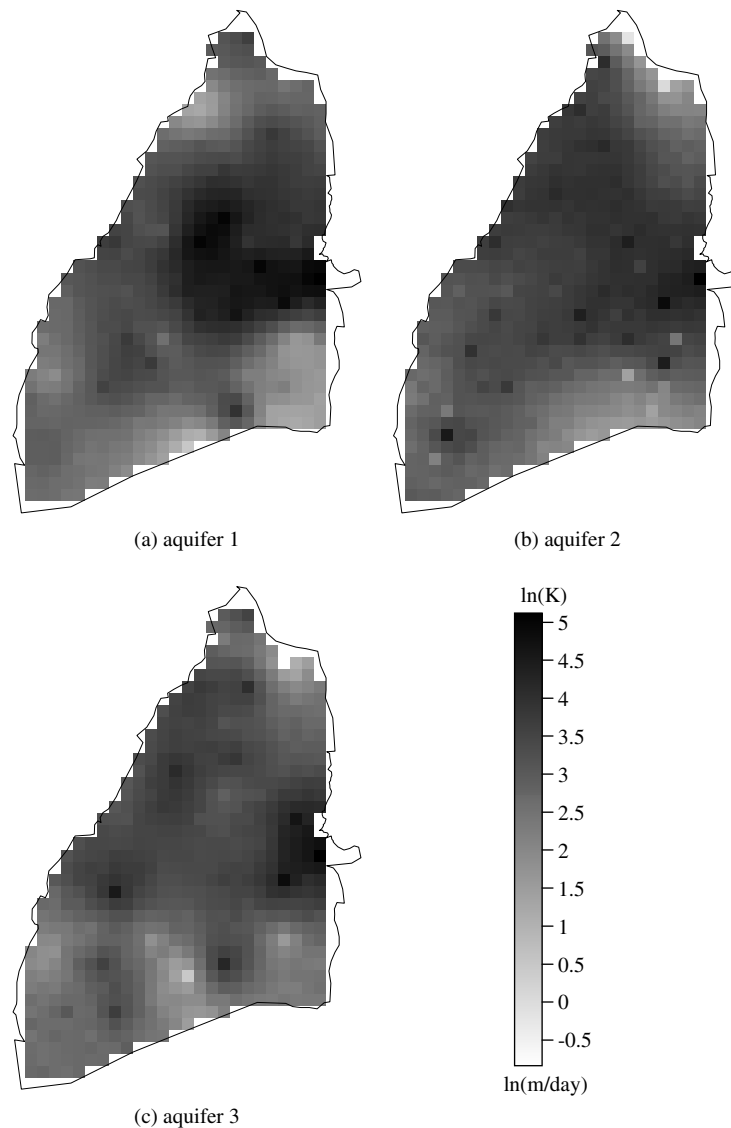


Figure 7. Mean SGS realization of $\ln(K)$ in (a) aquifer 1, (b) aquifer 2 and (c) aquifer 3

Cressie, 1985). The fitting results demonstrate that the spherical model fits better than the others in aquifers 1 and 2, and that the exponential model fits better than the others in aquifer 3. The spherical and exponential models are defined as follows (Isaaks and Srivastava, 1989)

$$\text{Spherical model : } \begin{cases} \gamma(h) = c_0 + c \left[1.5 \left(\frac{h}{a} \right) - 0.5 \left(\frac{h}{a} \right)^3 \right], & h \leq a \\ \gamma(h) = c_0 + c, & h > a \end{cases} \quad (2)$$

$$\text{exponential model : } \gamma(h) = c_0 + c \left[1 - \exp \left(-\frac{3h}{a} \right) \right] \quad (3)$$

where c_0 is the nugget effect, c is the sill, a is the range and h is the lag distance. The three parameters (c_0 , c and a) are determined through fitting experimental variograms. Figure 5 presents the fitting parameters of the models. The $\ln(K)$ variogram reveals no nugget effect in aquifer 1, but nugget effects of 0.3 and 0.1 $(\ln(\text{m/day}))^2$, 33% and 9% of the sill, in aquifers 2 and 3, respectively. The sill of 2 $(\ln(\text{m/day}))^2$ in aquifer 1 is the largest, and those of aquifers 2 and 3 are 0.9 and 1.1 $(\ln(\text{m/day}))^2$, respectively. Aquifer 2 has the longest range, at 36 km, and aquifer 1 has the shortest range, at 28 km. The number of pairs in the experimental variogram is insufficient to represent properly the spatial variation at long correlation lengths (above 40 km). In the experimental variograms, the data within 40 km are used to validate the assumption of stationary.

Simulation error for K realizations using OK

$\ln(K)$ realization generated by OK visually smoothes out spatial variations and yields rather continuous distributions, as shown in Figure 6. Table II analyses the descriptive statistics concerning OK realizations. The OK realization overestimates low values and underestimates high values as compared with measured data. A smoothing effect is insignificant in aquifer 1 and significant in aquifer 2. The maximum and minimum values of the OK realization are 4.47 and 0.83 $\ln(\text{m/day})$ in aquifer 2, respectively. Following an exponential transformation, the smoothing effect is illustrated further. The maximum estimated K is 87.4 m/day, which is considerably smaller than the maximum measured value of 167.7 m/day. The minimum estimated K is 2.3 m/day, which is larger than the minimum measured value of 0.8 m/day.

The OK realization of K field was imported into MODFLOW for simulation. Other parameters and simulation conditions were fixed as those used in the model calibration. The computed SAMSE is 15.1 m for the K field realized by OK.

Simulation error for K realizations using SGS

Mean SGS realization. Eggleston *et al.* (1996) suggested that 50–400 SGS realizations were required to obtain a statistically stable mean realization. This work thus adopted the mean of 100 reproduced realizations. The mean SGS realization of $\ln(K)$ are more irregular and heterogeneous than the OK realization, as displayed in Figure 7, especially in aquifer 2. The descriptive statistics of the mean SGS realization are distributed within the same range of measured data, have a smaller standard deviation and are less skewed than those of the OK realization (Table II). Notably, the nugget effect does not influence the mean SGS realization of $\ln(K)$ to reproduce extreme values of measured data. The mean SGS realizations of the K field are inputted to MODFLOW for simulation. The computed SAMSE is 15.5 m, which is slightly larger than that of the OK realization.

Individual SGS realization. The individual SGS realization of $\ln(K)$ statistically varies greatly because mean values of 100 realizations range from 2.04 to 3.52 and the corresponding standard deviations of 100

Table II. Descriptive statistics of estimated $\ln(K)$ realizations

Algorithm	Aquifer	Mean ($\ln(\text{m/day})$)	Standard deviation ($\ln(\text{m/day})$)	Skewness	Minimum ($\ln(\text{m/day})$)	Maximum ($\ln(\text{m/day})$)
OK	1	2.96	0.94	−0.71	−0.55	5.04
	2	3.09	0.80	−0.91	0.83	4.47
	3	3.10	0.77	−0.67	0.32	4.83
Mean SGS	1	2.95	0.86	−0.32	−0.84	5.12
	2	3.10	0.74	−0.80	−0.21	5.12
	3	3.06	0.57	−0.32	−0.45	5.12
Individual SGS	1	3.31	1.40	−0.73	−0.84	5.12
	2	3.20	1.04	−1.09	−0.21	5.12
	3	3.28	1.08	−1.36	−0.45	5.12

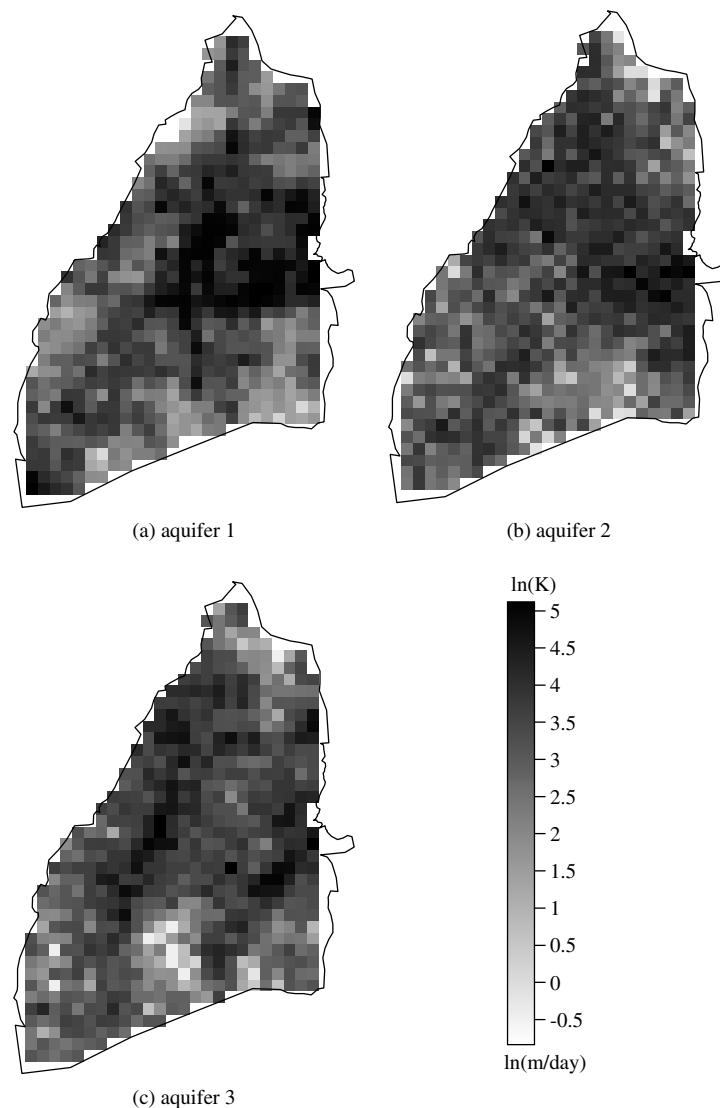


Figure 8. Individual SGS realization of $\ln(K)$ with the smallest SAMSE among 100 realizations in (a) aquifer 1, (b) aquifer 2 and (c) aquifer 3

realizations range from 0.8 to 1.92. One hundred SGS realizations of K are individually utilized as inputs to MODFLOW for simulation. The computed SAMSEs range from 18.4 to 38.4 m, with a mean of 27.1 m over 100 realizations. The individual SGS realization of the smallest SAMSE is of interest and its spatial distribution and statistical properties are analysed. The SGS realization of the smallest SAMSE displays high spatial variability and heterogeneity in the estimated $\ln(K)$ fields (Figure 8), while its statistical properties are similar to those of the measured data in the three aquifers (Tables I and II).

Statistical stability of SGS realizations and simulated heads

This work stochastically considered 100 sets of the K realizations and the corresponding simulated heads. A statistical stability analysis involved calculating maximum differences between mean values and standard deviations of randomly averaged N realizations. Figure 9a and b plots mean values and standard deviations

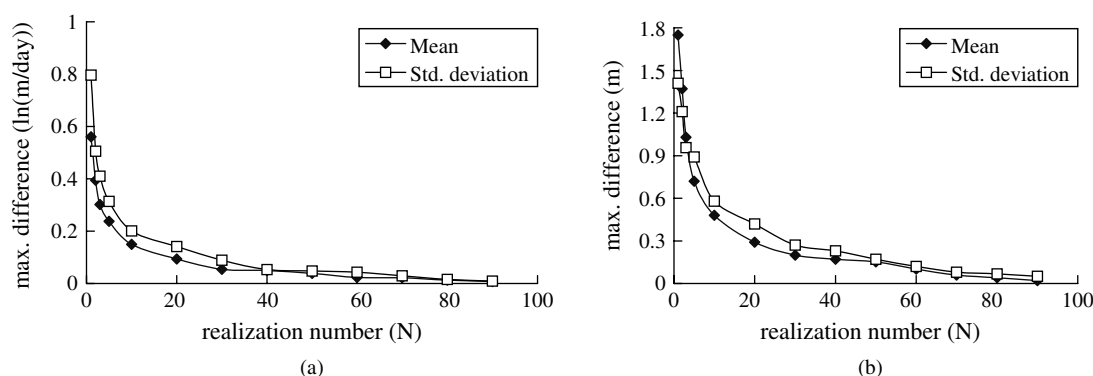


Figure 9. Test for the statistical stability of (a) reproducing $\ln(K)$ realizations and (b) corresponding simulated heads

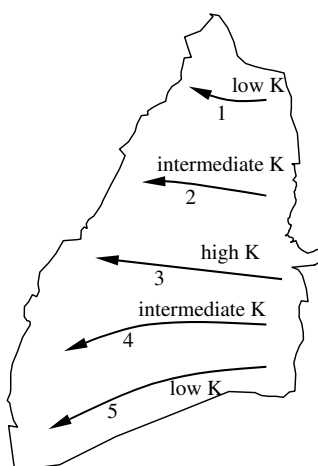


Figure 10. Five proposed major flow paths of groundwater in the Choushui River alluvial fan

Table III. The SAMSE along five major flow paths in the Choushui River alluvial fan

Flow path	1	2	3	4	5
Including zones for aquifer 1 ^a	1,2,3	4,5,6	7,9,11	8,10,12,15	13,14,15
Average K , m/day	24.4	41.0	65.9	31.2	21.1
Minimum K , m/day	0.6	3.4	18.0	3.9	0.4
Maximum K , m/day	60.9	67.4	167.3	106.7	52.5
SAMSE of OK, m	2.75	2.44	3.96	3.99	3.88
SAMSE of mean SGS, m	2.62	2.63	3.84	4.65	3.36
SAMSE of individual SGS, m	4.56	2.63	2.69	6.07	4.35

^a The zones included for aquifers 2 and 3 are in the same plan locations of aquifer 1, as shown in Figure 3.

for reproducing $\ln(K)$ and simulated heads, against the averaged N realization number. The curves markedly drop when the averaged number is 20. The maximum differences between the mean values and standard deviations of the averaged realizations are below 6% of the corresponding initial values ($N = 1$) when the average number is 70 and are below 3% of the corresponding initial values when the average number reaches

90. The result demonstrates that the 100 sets of $\ln(K)$ realizations and corresponding simulated heads exhibit sufficient statistical stability.

Local SAMSE along different flow paths

Simulation errors computed for measured and simulated heads frequently accumulate along the flow paths of groundwater, such as in vertically consecutive zones from the upper-layer aquifer to the lower-layer aquifer and horizontally consecutive zones from upstream to downstream. Figure 10 shows five major groundwater flow paths in the Choushui River alluvial fan along which local SAMSEs are evaluated. Flow paths 1 and 5 show low K fields and flow path 3 denotes a high K field, which include extreme values of K in the area of

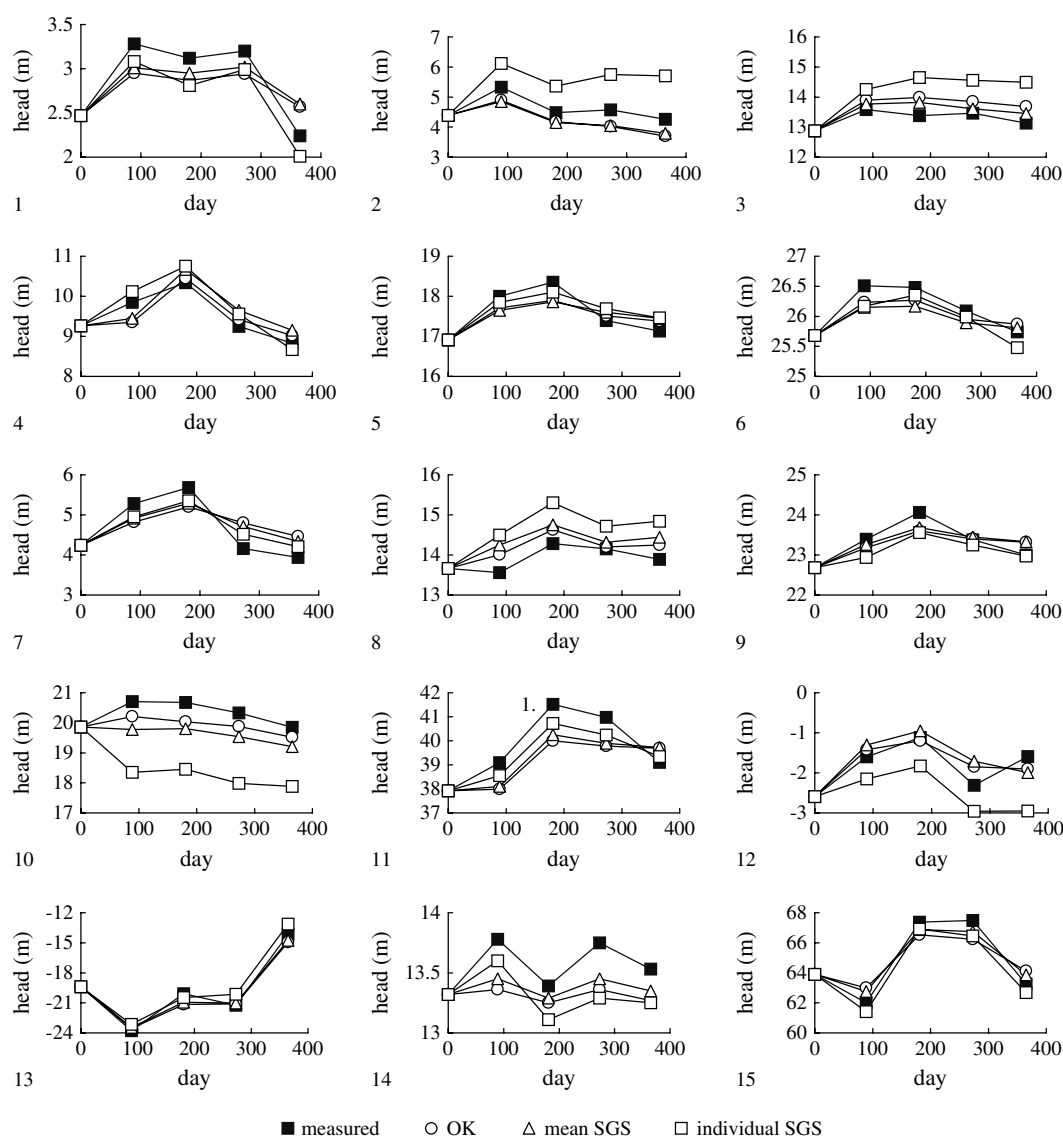


Figure 11. Comparison of historically measured heads and simulated heads of different K realizations in aquifer 1. The number of monitoring wells is consistent with their sequence in Figure 3

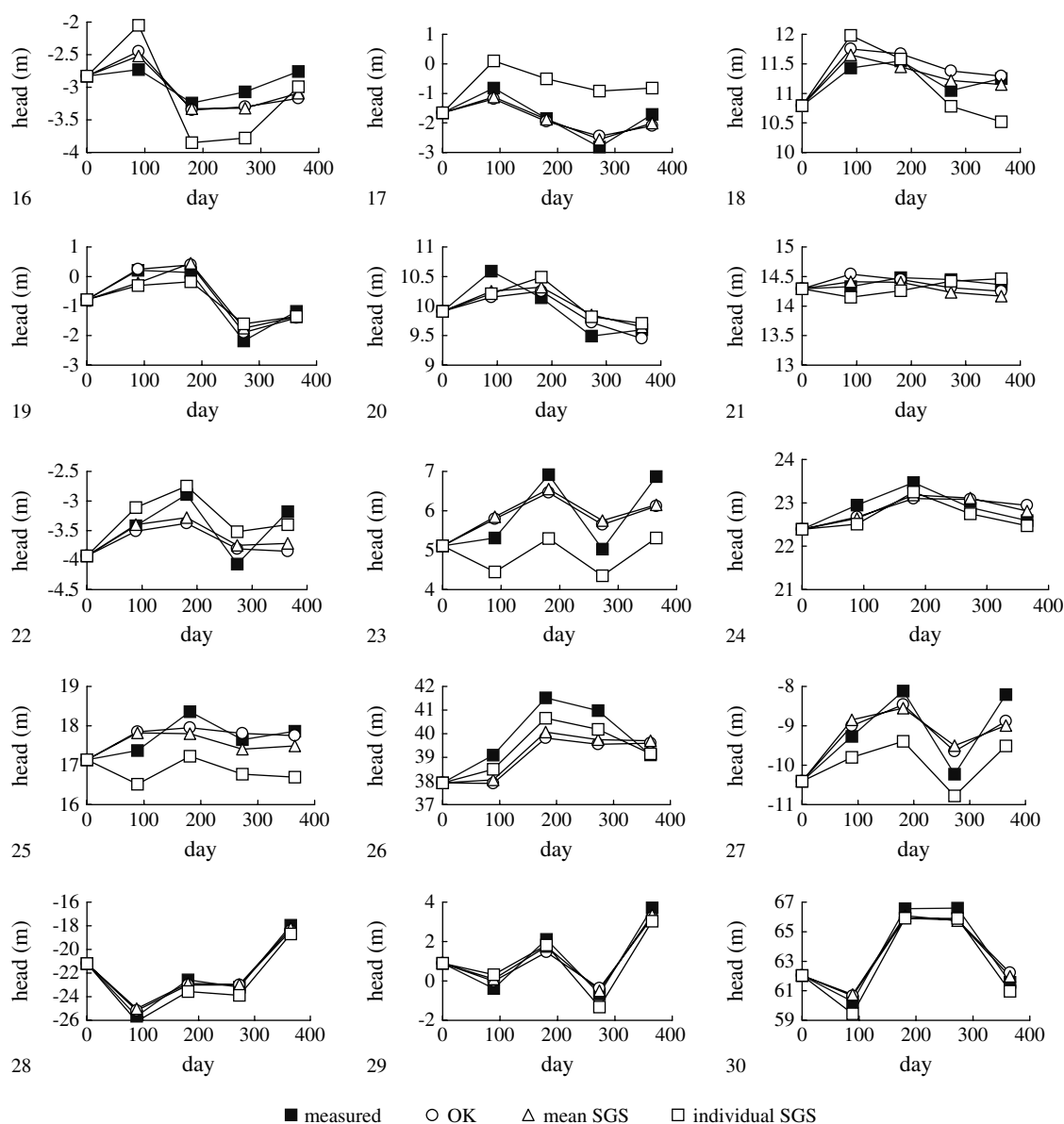


Figure 12. Comparison of historically measured heads and simulated heads of different K realizations in aquifer 2. The number of monitoring wells is consistent with their sequence in Figure 3

study (Table III). Flow paths 2 and 4 represent intermediate K fields indicating the average value of K in the study area.

The mean SGS realization exhibits small local SAMSEs along flow paths 1 and 5 with low K fields, and the individual SGS realization yields a small local SAMSE along flow path 3 with a high K field. The OK realization exhibits small local SAMSEs along flow paths 2 and 4 with intermediate K fields (see Table III). The simulated result indicates that the mean and individual SGS realizations adequately preserve high and low measured data, and hence have small local SAMSEs in flow paths 1, 3 and 5. The OK realization, which smoothes out extreme values and preserves intermediate measured K data, has small local SAMSEs

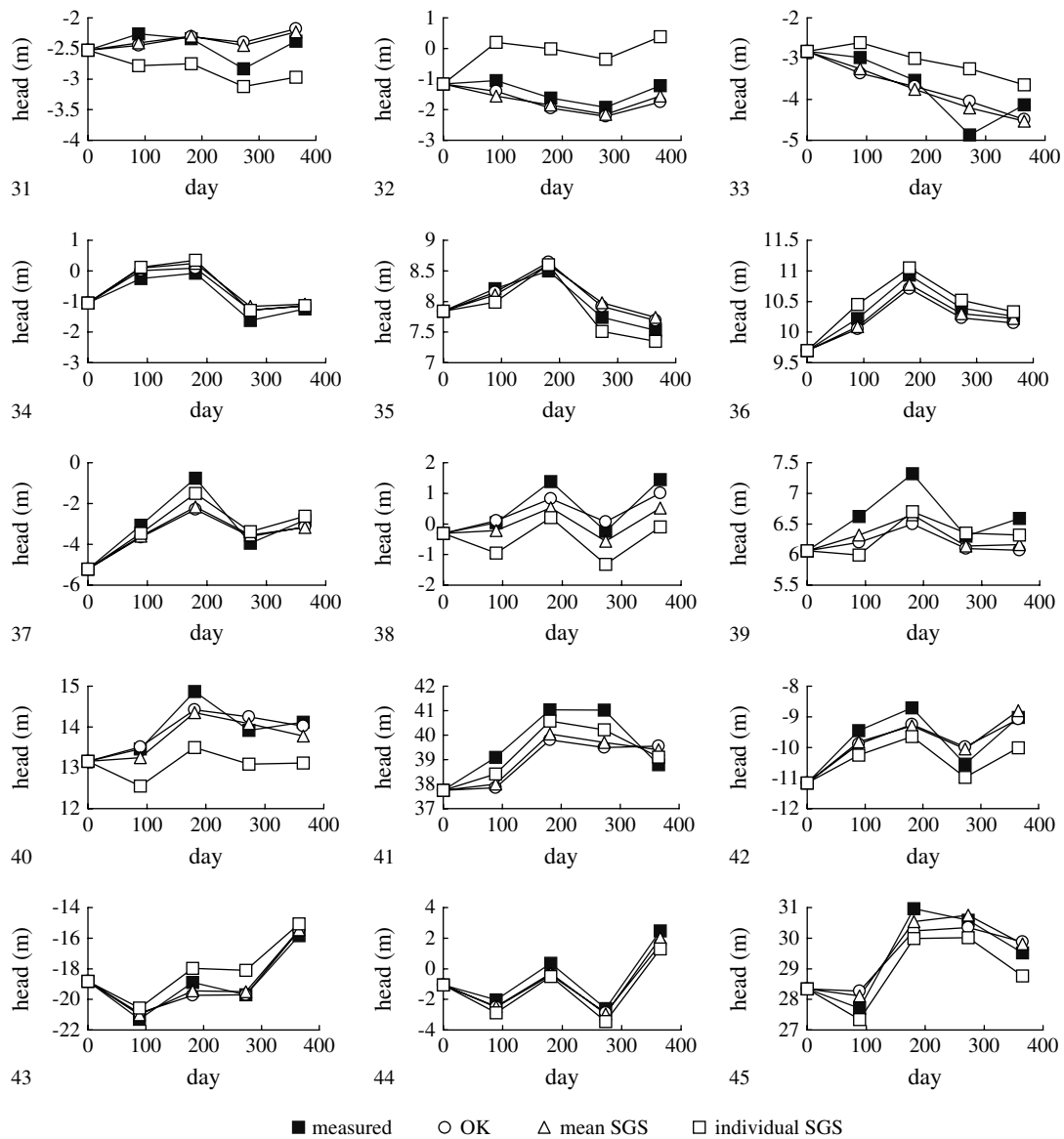


Figure 13. Comparison of historically measured heads and simulated heads of different K realizations in aquifer 3. The number of monitoring wells is consistent with their sequence in Figure 3

in flow paths 2 and 4. A slowed flow through low K fields causes heads to ascend in upstream regions and to descend in downstream regions. Meanwhile, an accelerated flow through high K fields causes heads to descend in upstream regions and to ascend in downstream regions. Consequently, highly varied K fields yield a few extreme values in the flow fields, and lead to large calibration errors. Figures 11, 12 and 13 compare historically measured heads and simulated heads using different K realizations. The fluctuations of simulated heads using the individual SGS realization of K are large, whereas the OK realization of K smoothes the head variations and reduces simulated errors.

SUMMARY AND CONCLUSIONS

This work adopted OK and SGS approaches to examine the spatial distribution and variability of K in the Choushui River alluvial fan, Taiwan. The OK realization smoothed out spatial variability and extreme measured values of K . The mean SGS realization yields moderate spatial variability, especially for spatial correlation structures that exhibit a nugget effect. The individual SGS realization shows high spatial variability and statistically extreme variation because of the wide ranges of the mean and standard deviation.

The OK algorithm not only provides an excellent spatial estimate but also reduces global simulated errors by smoothing spatial variability associated with a deterministic groundwater flow model. Thus, the OK realization of K can be applied to perform deterministic inverse calibration. Simulation errors increase with the spatial variability of K . The mean SGS realization yields a slightly higher global simulated error than OK. The mean SGS method is recommended for constructing K realizations when an application involves extremely varied parameters and small calibration errors, as in deterministic risk assessment associated with the movement of contaminants. A deterministic inverse problem is conducted to find a best parameter distribution with the smallest calibration error. The individual SGS realization cannot easily determine a regional K distribution with small simulation errors. The flow simulation obtained using an individual SGS realization of K complicates flow fields yielding large fluctuations of simulated heads in highly heterogeneous aquifers. Therefore, this approach is only appropriate for studying stochastic processes, revealing the spatial uncertainty of highly heterogeneous aquifer parameters. Furthermore, both mean SGS and individual SGS realizations can preserve extreme measured data and are less influenced by smoothing effects than OK. The OK realization yields small local simulation errors in the measured K field of moderate magnitude, whereas the SGS realizations have small local simulation errors in the measured K fields with high and low values.

Additionally, the SAMSEs analysis indicates that the OK and mean SGS realizations of K outperform the upscaling K in the baseline model simulation. The result confirms that a finely discretized grid of parameters estimated by geostatistical methods provides more accurate simulation results than zonal parameter upscaling.

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